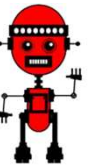
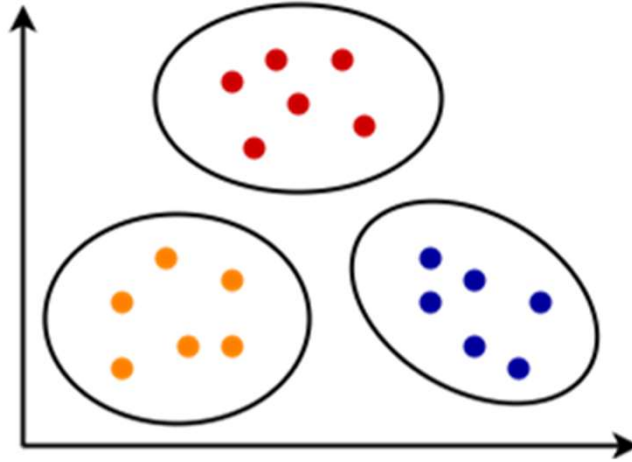


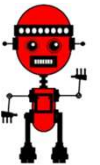
K-means Clustering Machine Learning Algorithm



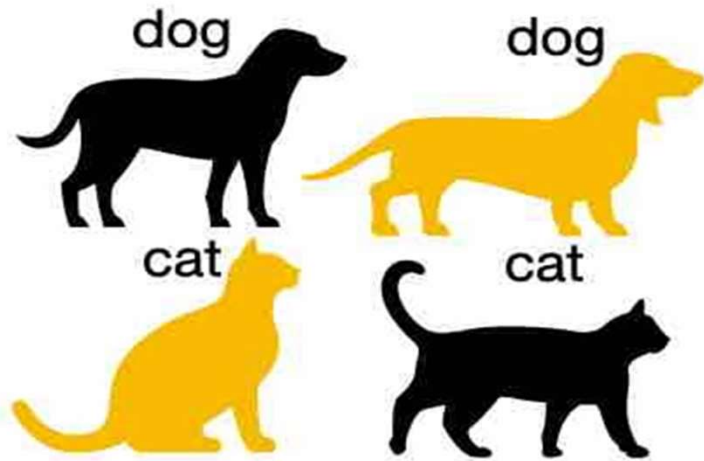
CONTENT



- Concept of Unsupervised Machine Learning Algorithm
- Concept of Clustering and Centroid
- How does K-means algorithm work ?
- How to choose best value for K ?

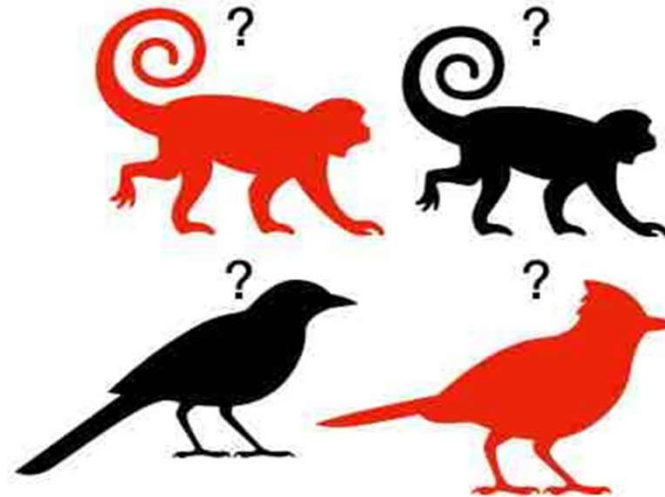


Supervised Machine Learning Algorithm



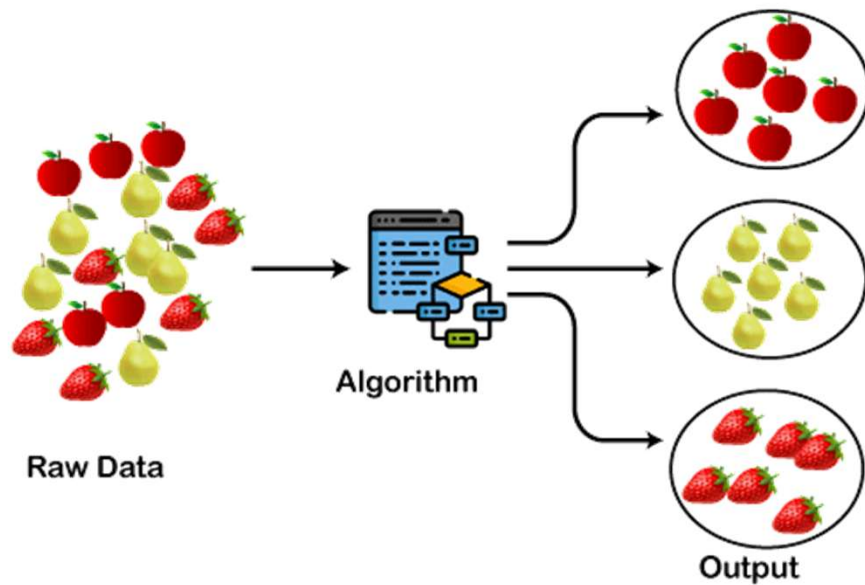
Labelled data

Unsupervised Machine Learning Algorithm

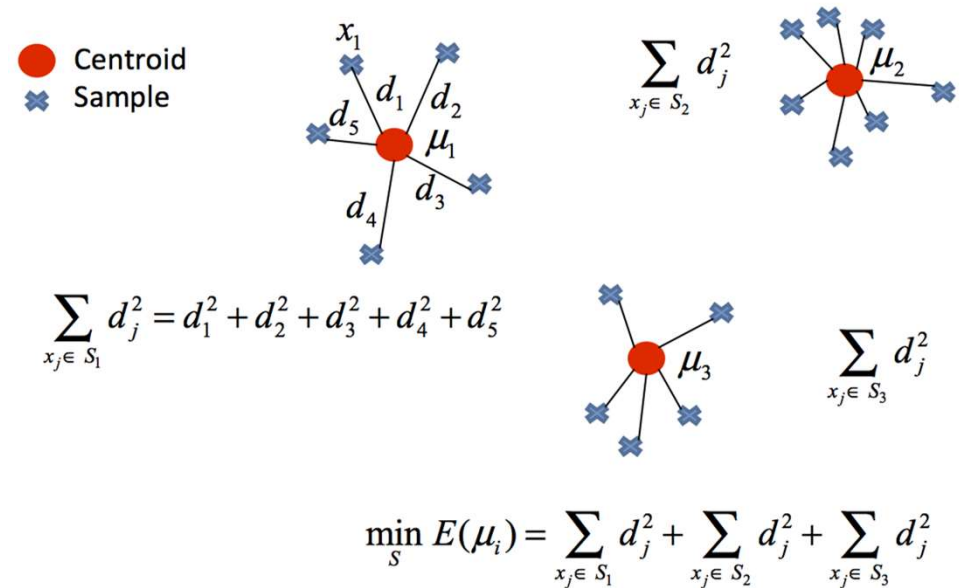


Unlabelled data

Clustering - Group similar objects

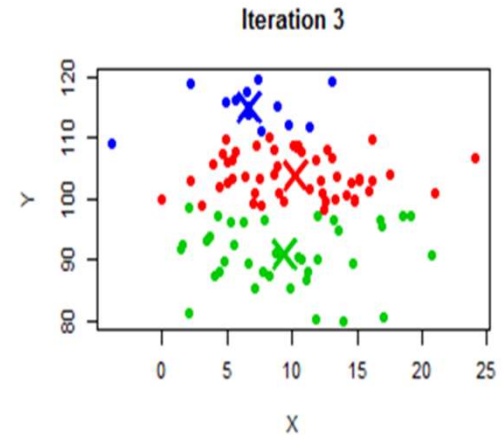
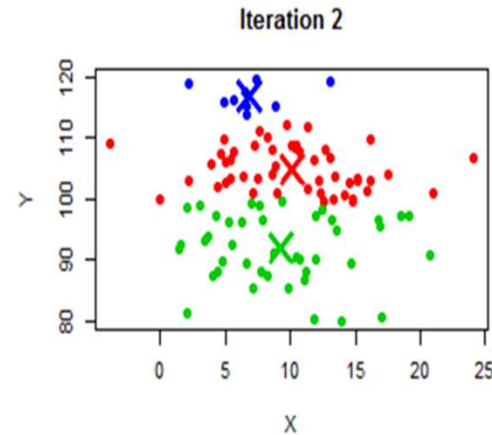
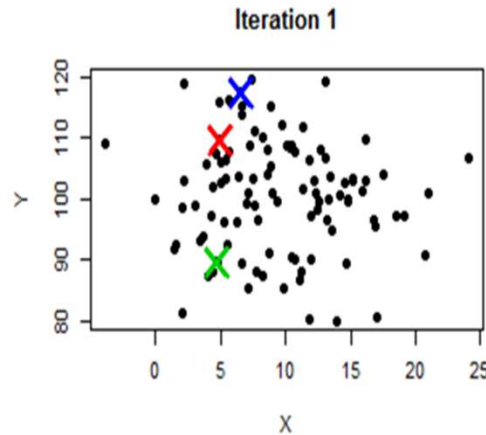


Centroid - Arithmetic mean of data points assigned to the cluster

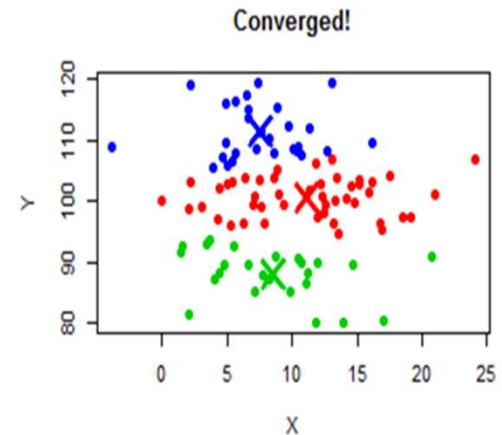
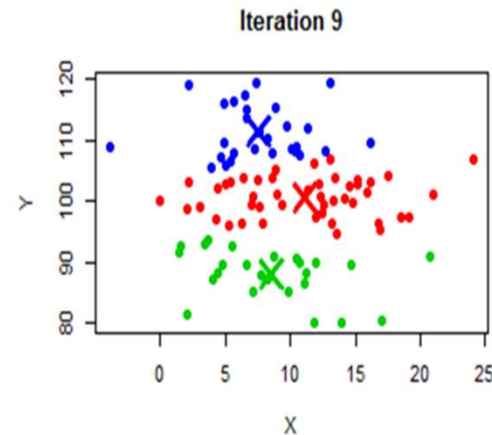
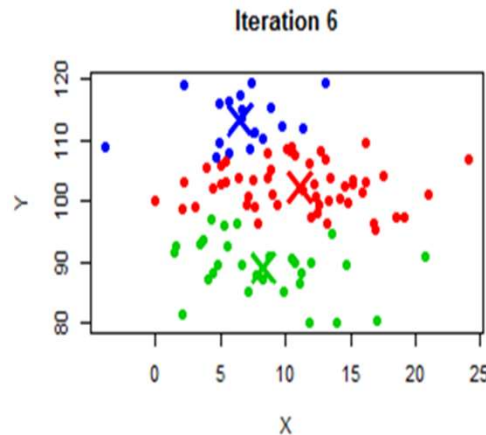


How does k-means algorithm work?

i) We randomly initialize the K starting centroids. Each data point is assigned to its nearest centroid.



ii) The centroids are recomputed as the mean of the data points assigned to the respective cluster.



iii) Repeat steps 1 and 2 until we trigger our stopping criteria.

How to choose best value for K ?

